sequent years, introduced policies to facilitate the drone industry's growth. These policies have focused on promoting intellectual property development, manufacturing, and exports. By easing regulations, providing financial incentives, and putting a hold on importing foreign drones, the government has created a conducive environment for Indian drone companies to thrive.

Recent government initiatives on drones:

- Drone policy liberalised via Drone Rules 2021
- Production Linked Incentive (PLI) scheme with 20% financial incentive
- Unmanned traffic management (UTM) policy framework published
- Pilot projects conducted across India for agriculture, healthcare, defence, etc
- SOP for spraying of pesticides & nutrients launched by the Ministry of Agriculture
- Subsidy announced by the Ministry of Agriculture for Kisan drones
- 100 Kisan drones launched by Prime Minister of India to promote the use of drones
- 1000-drone light show performed on Rashtrapati Bhawan
- DroneShakti Mission announced by India's Finance Minister
- Import ban on drones and knocked-down kits to promote domestic manufacturing

The Central government's vision is to make India a global hub for drones by 2030.

Drone Rules 2021 framework

The Drone Rules 2021 in India are structured around several crucial pillars to nurture a thriving drone ecosystem. Firstly, they prioritise simplifying the business environment for drone-related activities, reducing bureaucratic barriers and fostering innovation. Secondly, these rules

	Safety	Security		
Drone	Type Certification by QCI approved Certification body	Drone Registration	Insurance	No permission for flying in green zone
Drone Pilot	Remote Pilot Training by DGCA Approved Drone Training Schools	Remote Pilot Certificate		Permission for flying in yellow and red zones

Fig. 2: The Drone Rules 2021 advancing drone adoption nationwide

promote the domestic drone industry by prohibiting certain imports, bolstering local manufacturing, and enhancing national self-reliance in drone technology. Additionally, the rules incorporate financial incentives through the Production-Linked Incentive (PLI) scheme, encouraging local drone production and generating employment opportunities. Lastly, the government plays a pivotal role as a drone market participant, supporting the industry and demonstrating its commitment to integrating drone technology into critical sectors like agriculture, infrastructure, and public safety. The Drone Rules 2021 in India aim to facilitate business, boost the domestic drone industry, offer financial incentives, and establish the government as a critical driver in advancing drone technology adoption nationwide. It is depicted in Fig. 2.

Drone cybersecurity: India's path to secure skies

In the high-flying world of drones, the buzz isn't just about their capability to soar, capture breathtaking views, or deliver packages with pinpoint accuracy. It's also about ensuring these unmanned marvels are invulnerable to cyber threats. As India's skies become increasingly crowded with drones, how equipped are we in the face of evolving cybersecurity challenges? While drones bring unprecedented opportunities, they are not without their vulnerabilities. The fear isn't

Manufacturers	120+
Service Providers	150+
Training Schools	50+
Drone Pilots	50,000+
Skilled Professionals	10,000+

Fig. 3: The Indian drone industry

just about a drone falling out of the sky but the potential cyber-attacks on the vast troves of data they collect or malicious interference in their operation.

The concerns are valid, especially when considering the array of applications for drones, from military to civilian uses. Ensuring that drones and the data they collect remain secure is paramount. The Indian government, while championing the cause of drone innovation, is also taking active steps to ensure drone cybersecurity. The current drone-type certification scheme already mandates a series of security checks. This includes measures such as

Power on self-test (POST). Ensuring the integrity of drone systems each time they are powered on

Bootloader log requirements. Allowing only authenticated firmware to be loaded onto the drone systems

Password protection. Safeguarding unauthorised access to the drone's operational controls

Tamper detection and protection. Putting mechanisms in place to identify and prevent unauthorised physical and digital interference with the drone

While these measures provide a foundation for cybersecurity, it's evi-